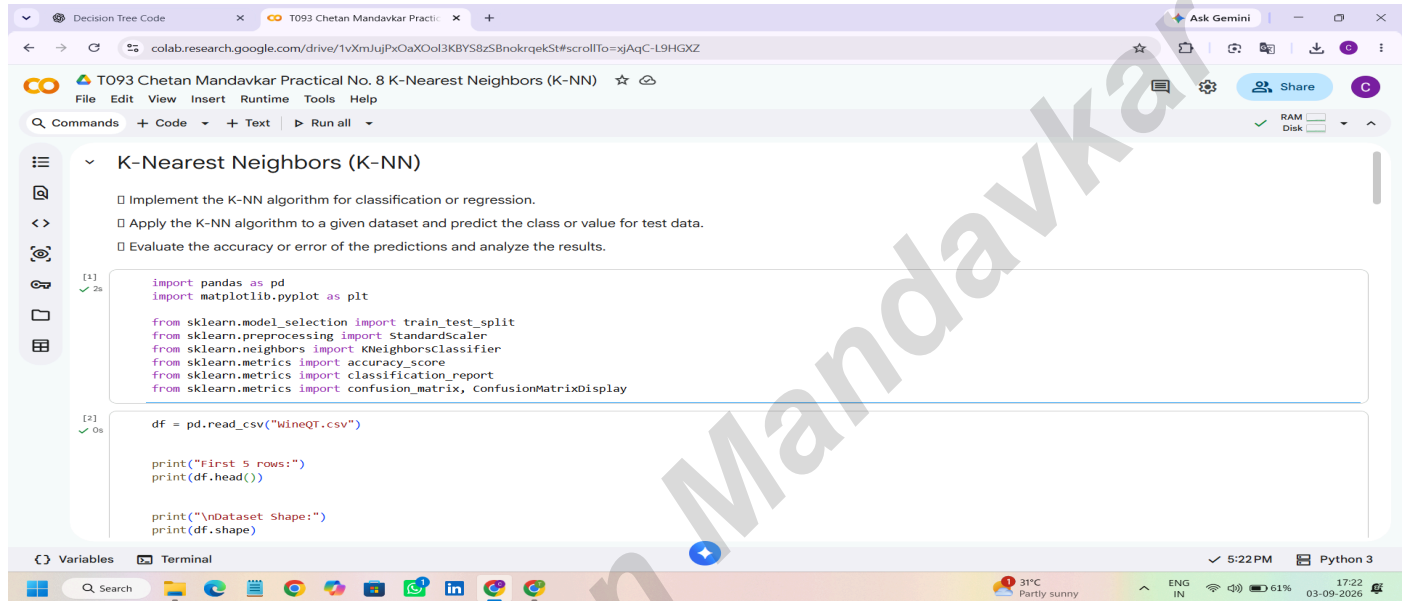


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Aim:- K-Nearest Neighbors (K-NN)

Implement the K-NN algorithm for classification or regression. Apply the K-NN algorithm to a given dataset and predict the class or value for test data. Evaluate the accuracy or error of the predictions and analyze the results.



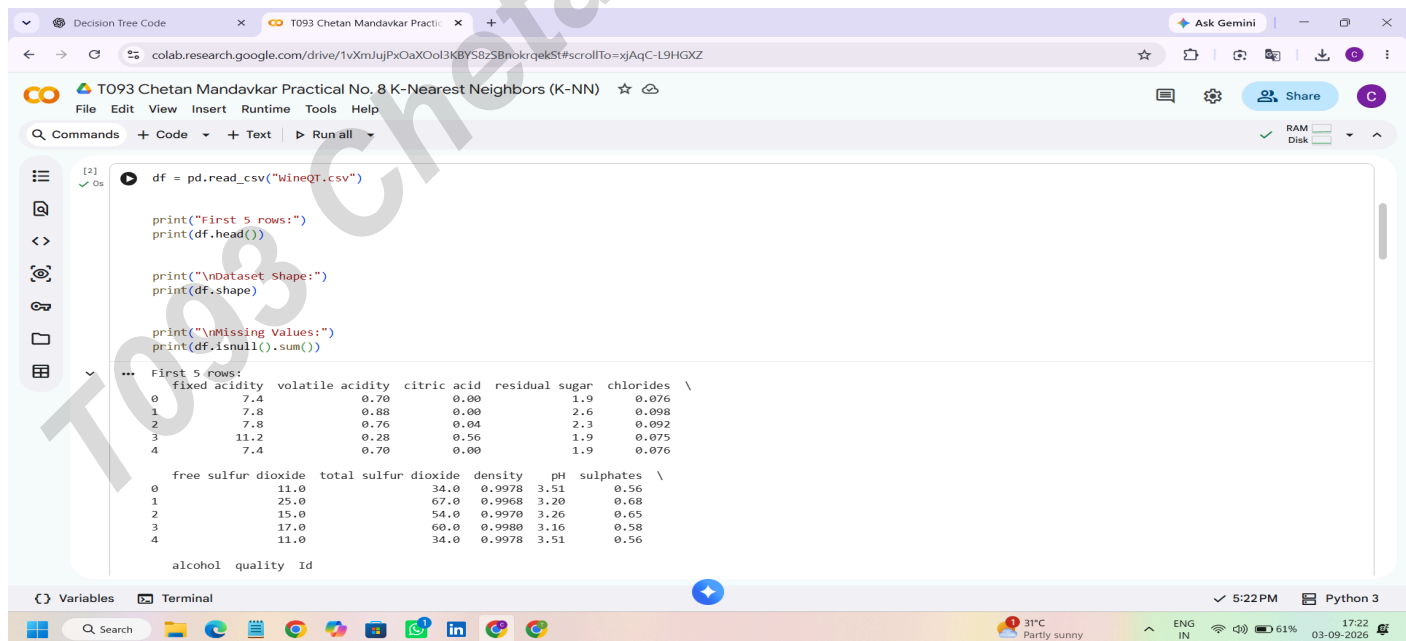
```
[1] import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

[2] df = pd.read_csv("wineQT.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)
```



```
[2] df = pd.read_csv("wineQT.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nMissing Values:")
print(df.isnull().sum())

... First 5 rows:
fixed acidity volatile acidity citric acid residual sugar chlorides \
0 7.4 0.70 0.00 1.9 0.076
1 7.8 0.88 0.00 2.6 0.098
2 7.8 0.76 0.04 2.3 0.092
3 11.2 0.28 0.56 1.9 0.075
4 7.4 0.70 0.00 1.9 0.076

free sulfur dioxide total sulfur dioxide density pH sulphates \
0 11.0 34.0 0.9978 3.51 0.56
1 25.0 67.0 0.9968 3.20 0.68
2 15.0 54.0 0.9970 3.26 0.65
3 17.0 60.0 0.9980 3.16 0.58
4 11.0 34.0 0.9978 3.51 0.56

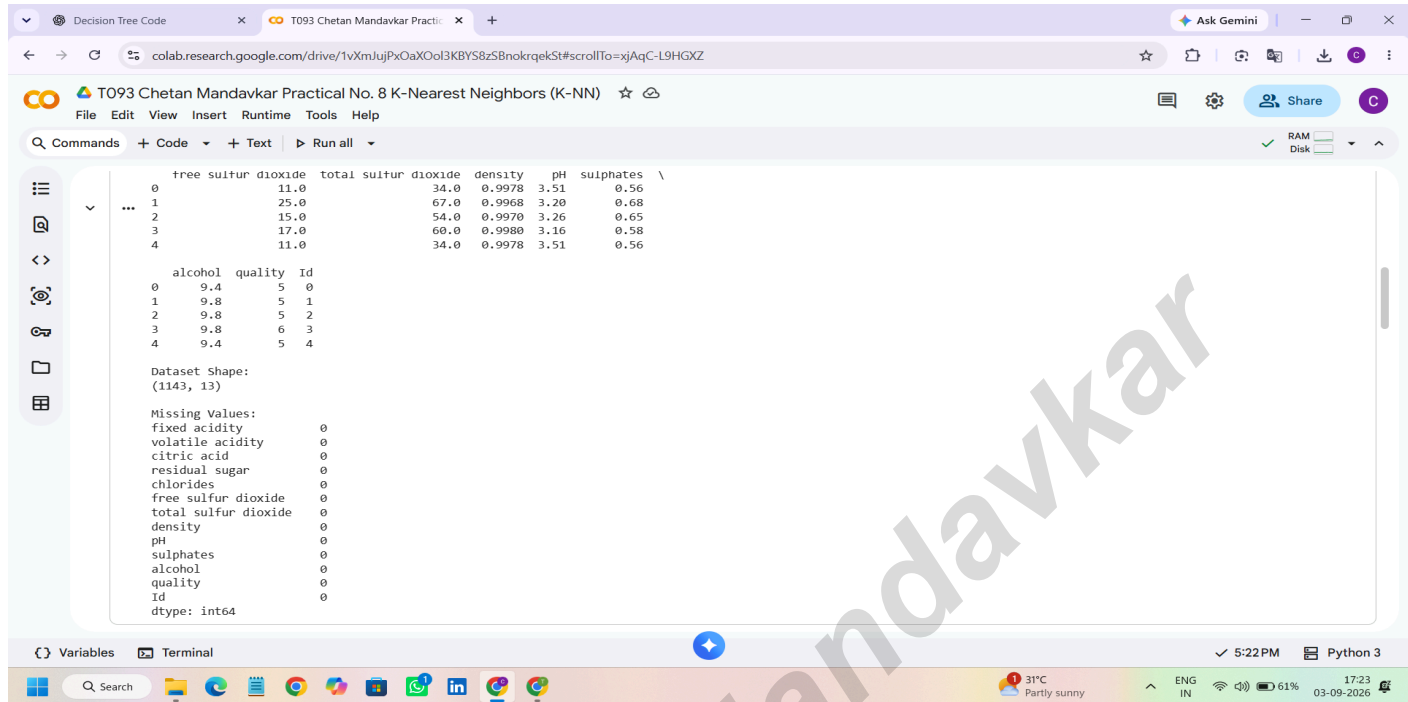
alcohol quality Id
```

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Decision Tree Code x T093 Chetan Mandavkar Practic x + Ask Gemini

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T093 Chetan Mandavkar Practical No. 8 K-Nearest Neighbors (K-NN) ☆ ☁

File Edit View Insert Runtime Tools Help

Commands + Code + Text ▶ Run all

```
tree sulfur dioxide total sulfur dioxide density pH sulphates \
0 11.0 34.0 0.9978 3.51 0.56
1 25.0 67.0 0.9968 3.20 0.68
2 15.0 54.0 0.9970 3.26 0.65
3 17.0 60.0 0.9980 3.16 0.58
4 11.0 34.0 0.9978 3.51 0.56

alcohol quality Id
0 9.4 5 0
1 9.8 5 1
2 9.8 5 2
3 9.8 6 3
4 9.4 5 4

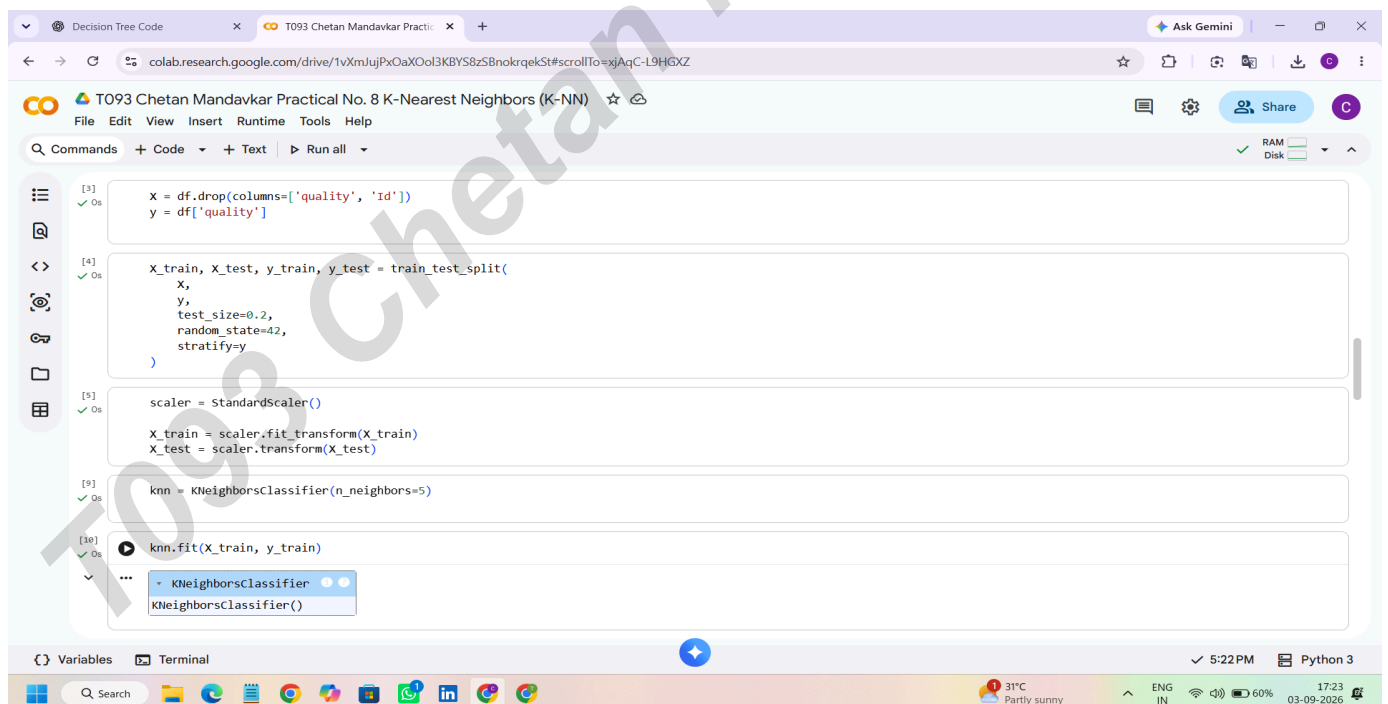
Dataset Shape:
(1143, 13)

Missing Values:
fixed acidity 0
volatile acidity 0
citric acid 0
residual sugar 0
chlorides 0
free sulfur dioxide 0
total sulfur dioxide 0
density 0
pH 0
sulphates 0
alcohol 0
quality 0
Id 0
dtype: int64
```

Variables Terminal

5:22 PM Python 3

31°C Partly sunny ENG IN 61% 17:23 03-09-2026



Decision Tree Code x T093 Chetan Mandavkar Practic x + Ask Gemini

colab.research.google.com/drive/1vXmJujPxOaXOol3KBYS8zS8nokrqekSt#scrollTo=xjAqC-L9HGxZ

T093 Chetan Mandavkar Practical No. 8 K-Nearest Neighbors (K-NN) ☆ ☁

File Edit View Insert Runtime Tools Help

Commands + Code + Text ▶ Run all

```
[3] ✓ Os
x = df.drop(columns=['quality', 'Id'])
y = df['quality']

[4] ✓ Os
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

[5] ✓ Os
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

[9] ✓ Os
knn = KNeighborsClassifier(n_neighbors=5)

[10] ✓ Os
knn.fit(X_train, y_train)
```

Variables Terminal

5:22 PM Python 3

31°C Partly sunny ENG IN 60% 17:23 03-09-2026

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The screenshot shows a Google Colab environment with the following code and output:

```
[11] ✓ Os y_pred = knn.predict(X_test)
[12] ✓ Os accuracy = accuracy_score(y_test, y_pred)
[13] ✓ Os print("\nAccuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

Output:

```
Accuracy: 0.5764192139737991
Accuracy Percentage: 57.64192139737992 %

Classification Report:
              precision    recall  f1-score   support

     3         0.00         0.00         0.00         1
     4         0.00         0.00         0.00         7
     5         0.68         0.68         0.68         97
     6         0.53         0.61         0.57         92
     7         0.42         0.34         0.38         29
     8         0.00         0.00         0.00         3

 accuracy          0.27         0.27         0.58        229
 macro avg          0.27         0.27         0.27        229
 weighted avg       0.55         0.58         0.56        229
```

The screenshot shows the same Google Colab environment with the following code and output:

```
[13] ✓ Os print("Accuracy Percentage:", accuracy * 100, "%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

Output:

```
Accuracy: 0.5764192139737991
Accuracy Percentage: 57.64192139737992 %

Classification Report:
              precision    recall  f1-score   support

     3         0.00         0.00         0.00         1
     4         0.00         0.00         0.00         7
     5         0.68         0.68         0.68         97
     6         0.53         0.61         0.57         92
     7         0.42         0.34         0.38         29
     8         0.00         0.00         0.00         3

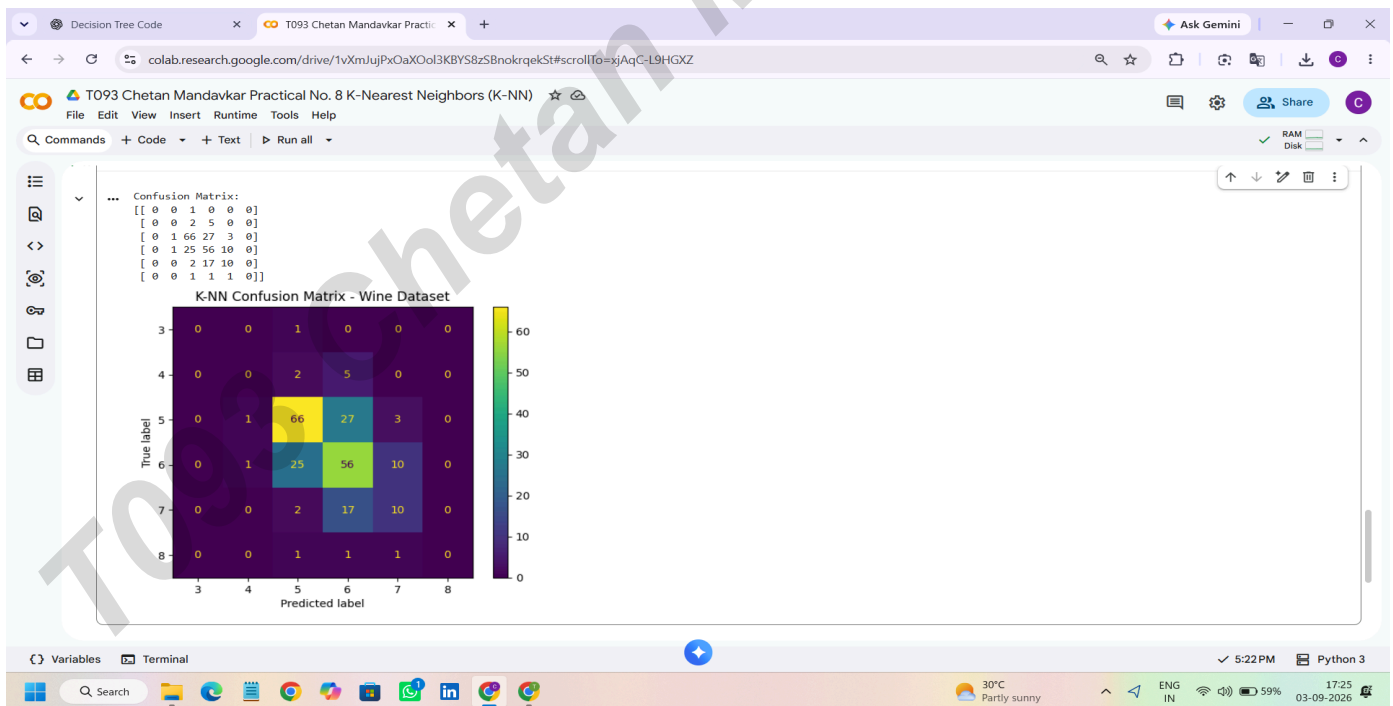
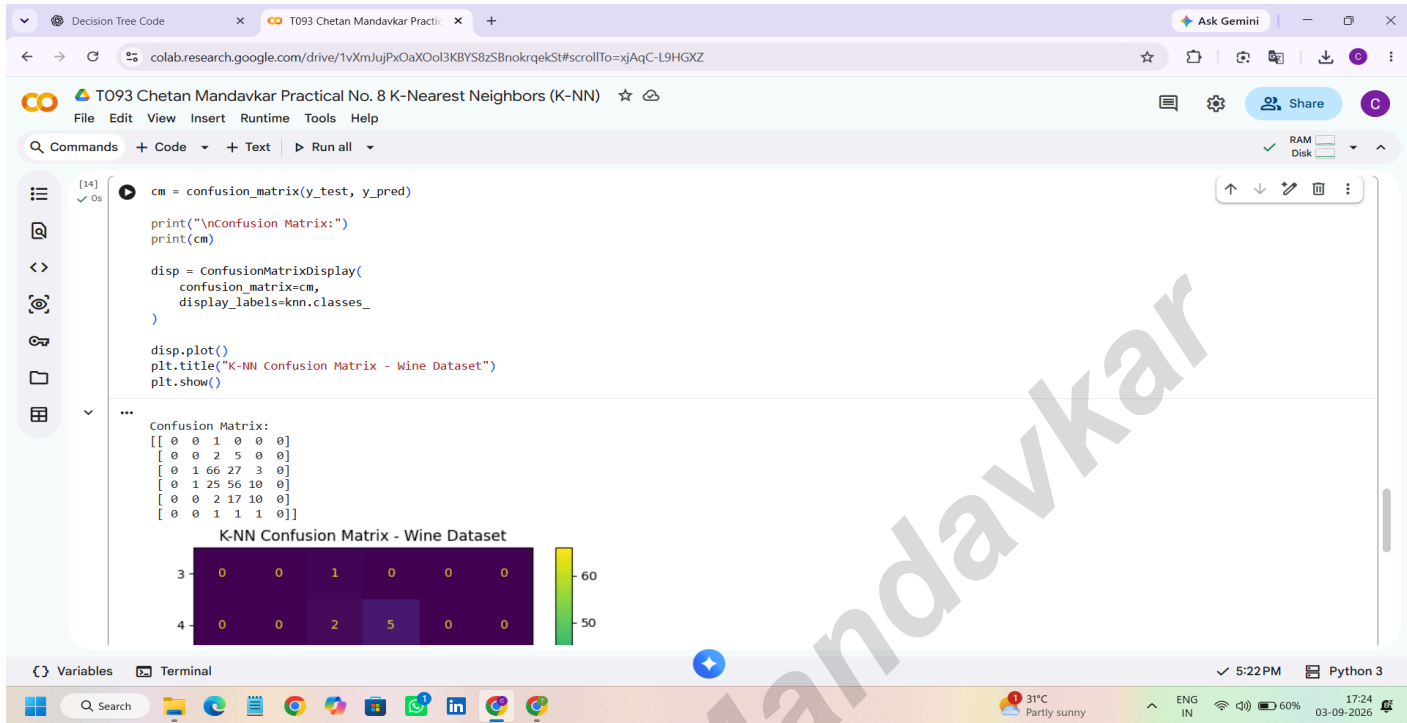
 accuracy          0.27         0.27         0.58        229
 macro avg          0.27         0.27         0.27        229
 weighted avg       0.55         0.58         0.56        229
```

Warnings:

```
/usr/local/lib/python3.13/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.13/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.13/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

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